

...Automatic Music Orchestration ...

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Abstract

The abstract paragraph should be indented 1.5 cm on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 INTRODUCTION

TODO: - What is orchestration

- Orchestration is more than instrumentation and instrument estimation (but you need to start somewhere)

2 STATE OF THE ART

TODO: - Focus on symbolic orchestration methods

- Frameworks: different tasks, how they go together, fully autonomous end-to-end vs co-creative. I think here we have a process in steps

- Distinguish from Orchidea and other sound imitation orchestration methods

- Present Gissel's Orchestration by Example (ObyE): orchestrate by learning from a model example, use tabular data representation, avoid data-hungry methods and use tree-based classification. Highlight advantages. Highlight problems addressed here.

3 METHODS

3.1 ObyE with multi-instrument and transpositions (Francesco)

ObyE provided a lightweight method to estimate the instrumentation of a piece. It does not require large training datasets (only one example file) and it does not need long training times. This method however only allows to assign existing notes to instruments, and lacks the possibility of transforming the existing note content or creating new notes. This is a fundamental aspect of professional human orchestration, that heavily rely on "rewriting" rather than distributing the music content.

TODO: Francesco: Continue this description

TODO: Add a scheme of the algorithm

TODO: Francesco: how can we make this better with additional features?

TODO: Francesco: how can we make this better now that we have a way to align the piano and orchestral pieces in the training dataset?

4 RESULTS

FM: « We are not completely happy with what comes out of the model but we find something interesting »

TODO: Report the accuracy, precision, recall, f1 data and figure. Compare models

5 DISCUSSION

TODO: Discuss what is learned.

FM: « I believe that the model is learning the "structure" of the example piece rather than learning something really about orchestration. Different example pieces produce very different tree structures. However, this can be used if we are careful to provide example pieces that resemble the piece that we want to orchestrate. »

TODO: Report some figures with different pieces to show how notes are classified based on the features.

TODO: Francesco: What would these figures become if we train on the entire dataset?

TODO: Francesco: can the alignment be improved by considering that some repetitions might be expanded in one version?

6 CONCLUSION

AUTHOR DECLARATIONS

Author declarations should include conflicts of interest, ethical issues, and comments on the use of AI.

Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references.

Do **not** include acknowledgments in the anonymized submission, only in the final paper. You can use the ack environment provided in the style file to automatically hide this section in the anonymized submission.

REFERENCES

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